Nonlinear Dynamics and Chaos

Comprehensive Lecture Notes

Based on the course following
Nonlinear Dynamics and Chaos (3rd Edition)
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These notes are self-contained and assume familiarity with multivariable calculus, linear algebra, and ordinary differential equations at the level of a mathematical methods of physics course.

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Part I
One-Dimensional Flows

1 Flows on the Line

1.1 First-Order ODEs and Autonomous Systems

The most general first-order ordinary differential equation (ODE) for a single real variable $x(t)$ is

$$\dot{x} = f(x, t), \tag{1}$$

where the overdot denotes $\mathrm{d}x \,/\, \mathrm{d}t$.

Definition 1.1 (Autonomous ODE)**.** An ODE of the form

$$\dot{x} = f(x) \tag{2}$$

is called **autonomous** because the right-hand side does not depend explicitly on time t. The function f specifies the *velocity* of the state x on the real line.

The system (2) is **linear** if $f(x) = ax + b$ for constants a, b, and **nonlinear** otherwise.

Given an initial condition $x(0) = x_0$, the solution $x(t)$ traces a trajectory on the real line. Because f depends only on x, the velocity at a given position x is the same regardless of when the particle passes through x. This is the hallmark of an autonomous system.

1.2 A Geometric Way of Thinking: The Phase Line

Rather than solving (2) analytically (which is often impossible for nonlinear f), we adopt a *geometric* approach.

Procedure.

1. Plot $f(x)$ versus x.
2. Where $f(x) > 0$, the flow moves to the *right* (increasing x); where $f(x) < 0$, the flow moves to the *left*.
3. Points where $f(x^*) = 0$ are at rest—they are called **fixed points** (or equilibria, critical points, stationary points).

Definition 1.2 (Fixed Point)**.** A point x^* is a **fixed point** of $\dot{x} = f(x)$ if $f(x^*) = 0$.

Definition 1.3 (Stability of a Fixed Point—Informal)**.** A fixed point x^* is

- **stable** (an "attractor") if nearby trajectories approach x^* as $t \to \infty$;
- **unstable** (a "repeller") if nearby trajectories move away from x^*.

Graphical rule. Looking at the graph of $f(x)$: a fixed point where f crosses from positive to negative (i.e., $f'(x^*) < 0$) is stable; a crossing from negative to positive ($f'(x^*) > 0$) is unstable.

Direction Field

For autonomous systems, the direction field has identical slope $f(x)$ at every point along a horizontal line $x = \text{const}$. Solution trajectories in the (t, x) plane are parallel translations of each other. The initial condition x_0 simply selects which trajectory the system follows.

1.3 Linear Stability Analysis

To make the stability criterion precise, we linearize near the fixed point.

Let x^* be a fixed point and set $u = x - x^*$ (a small perturbation). Then

$$\dot{u} = \dot{x} = f(x) = f(x^*) + f'(x^*)\,u + O(u^2).$$

Since $f(x^*) = 0$, dropping higher-order terms gives the *linearized equation*:

$$\dot{u} \approx f'(x^*)\,u. \tag{3}$$

This is a linear ODE with solution

$$u(t) = u_0\, e^{f'(x^*)\,t}. \tag{4}$$

Linear Stability Criterion (1D)

Let x^* be a fixed point of $\dot{x} = f(x)$.

- If $f'(x^*) < 0$, then x^* is **linearly stable**.
- If $f'(x^*) > 0$, then x^* is **unstable**.
- If $f'(x^*) = 0$, the test is **inconclusive**—higher-order terms must be examined.

Example 1.1 (Population Growth)**.** Consider $\dot{x} = rx(1 - x/K)$ (the logistic equation) with $r, K > 0$. The fixed points are $x^* = 0$ and $x^* = K$. Since $f'(x) = r - 2rx/K$, we have $f'(0) = r > 0$ (unstable) and $f'(K) = -r < 0$ (stable). The carrying capacity K is the stable equilibrium.

1.4 Existence and Uniqueness

Theorem 1.1 (Existence and Uniqueness)**.** *Consider the initial value problem $\dot{x} = f(x)$, $x(0) = x_0$. If $f(x)$ is continuous and $f'(x)$ is also continuous (i.e., $f \in C^1$) on an open interval containing x_0, then a solution $x(t)$ exists and is unique on some time interval $(-\tau, \tau)$ about $t = 0$.*

Remark 1.1. A key consequence is that trajectories on the real line *cannot cross* fixed points. This means the qualitative behavior is completely determined by the signs of $f(x)$ between consecutive fixed points.

1.5 Impossibility of Oscillations

Theorem 1.2. *For a first-order autonomous ODE $\dot{x} = f(x)$ on the real line with $f \in C^1$, oscillations (periodic solutions) are impossible.*

Proof. The velocity $f(x)$ at any point x is single-valued and time-independent. For the system to oscillate, x would have to return to a previously visited value, requiring $\dot{x}$ to change sign while passing through the same point—but $f(x)$ is fixed at each x. More precisely, by the uniqueness theorem, trajectories cannot cross, and on $\mathbb{R}^1$ there is no way to return to a starting point without crossing a fixed point, at which the solution would be trapped forever. □

1.6 Potentials

For one-dimensional flows, we can often think of the dynamics in terms of a "particle rolling downhill on a potential landscape."

Definition 1.4 (Potential Function)**.** Given $\dot{x} = f(x)$, define the **potential** $V(x)$ by

$$f(x) = -\frac{\mathrm{d}V}{\mathrm{d}x}, \qquad \text{i.e.} \qquad V(x) = -\int f(x)\,dx. \tag{5}$$

Then $\dot{x} = -V'(x)$: the system always moves in the direction of decreasing V. Stable fixed points correspond to local *minima* of V; unstable fixed points correspond to local *maxima*.

The potential decreases monotonically along trajectories:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = V'(x)\,\dot{x} = V'(x)\big[-V'(x)\big] = -\big[V'(x)\big]^2 \leq 0.$$

1.7 Solving Equations on the Computer: Euler's Method

In practice, many nonlinear ODEs cannot be solved in closed form. The simplest numerical scheme for $\dot{x} = f(x)$ is **Euler's method**:

$$x_{n+1} = x_n + f(x_n)\,\Delta t, \tag{6}$$

which approximates the solution by taking small steps of size Δt along the tangent line. More sophisticated methods (Runge–Kutta, adaptive step-size) are used in practice, but Euler's method captures the essential idea.

Example 1.2 (Separation of Variables and Finite-Time Blowup)**.** Consider $\dot{x} = 1 + x^2$ with $x(0) = 0$. By separation of variables:

$$\int \frac{dx}{1+x^2} = \int dt \quad \Longrightarrow \quad \arctan(x) = t + C.$$

With $x(0) = 0$: $C = 0$, so $x(t) = \tan(t)$. The solution blows up in finite time $t = \pi/2$.

This illustrates an important caveat: the existence and uniqueness theorem guarantees solutions only *locally* in time. If f grows fast enough, the solution can escape to infinity in finite time. However, for flows on the *line* where f is smooth and bounded, this cannot happen.

Example 1.3 (Exact Solution: $\dot{x} = x^2 - 1$)**.** Fixed points: $x^* = \pm 1$. Since $f'(x) = 2x$, $f'(-1) = -2 < 0$ (stable) and $f'(1) = 2 > 0$ (unstable). By partial fractions:

$$\int \frac{dx}{x^2-1} = \int \frac{dx}{(x-1)(x+1)} = \frac{1}{2}\int \left(\frac{1}{x-1} - \frac{1}{x+1}\right) dx = t + C.$$

Solving gives $x(t) = -\tanh(t - t_0)$ for trajectories between the fixed points, confirming the graphical analysis.

2 Bifurcations

2.1 Introduction

A **bifurcation** occurs when a small, smooth change in a parameter causes a qualitative (topological) change in the dynamics—for instance, fixed points may be created, destroyed, or change stability.

The parameter is traditionally denoted r. The key question is: *How does the phase portrait of $\dot{x} = f(x; r)$ change as r varies?*

A **bifurcation diagram** plots a characteristic feature of the solution (usually the fixed point values x^*) on the vertical axis versus the parameter r on the horizontal axis. Solid curves denote stable fixed points; dashed curves denote unstable ones.

2.2 Saddle-Node Bifurcation

The **saddle-node bifurcation** is the basic mechanism by which fixed points are created and destroyed.

Normal Form: Saddle-Node Bifurcation

$$\dot{x} = r + x^2. \tag{7}$$

Analysis.

- For $r < 0$: two fixed points at $x^* = \pm\sqrt{-r}$. Since $f'(x) = 2x$, the negative root is stable ($f' < 0$) and the positive root is unstable ($f' > 0$).
- At $r = 0$: the two fixed points coalesce into one *half-stable* fixed point at $x^* = 0$.
- For $r > 0$: no fixed points exist—$f(x) > 0$ everywhere so x increases without bound.

The bifurcation occurs at $r_c = 0$. Graphically, the two branches of the bifurcation diagram meet in a cusp.

Bottleneck / Ghost Effect

Just above the bifurcation ($r = 0^+$), even though no fixed point exists, the system slows dramatically near $x = 0$ because $f(x) \approx r$ is small. The time to pass through this "bottleneck" scales as $T_{\text{bottleneck}} \sim \pi/\sqrt{r}$ for $r \to 0^+$. This is sometimes called the *ghost* of the vanished fixed point.

2.3 Transcritical Bifurcation

In many physical situations, a fixed point exists for all parameter values but exchanges stability with another fixed point.

Normal Form: Transcritical Bifurcation

$$\dot{x} = rx - x^2 = x(r - x). \tag{8}$$

Analysis. Fixed points: $x^* = 0$ and $x^* = r$. Since $f'(x) = r - 2x$:

- $f'(0) = r$: origin is stable for $r < 0$, unstable for $r > 0$.
- $f'(r) = -r$: the fixed point $x^* = r$ is unstable for $r < 0$, stable for $r > 0$.

At $r = 0$ the two fixed points collide and **exchange stability**. Neither fixed point is destroyed; they simply swap roles.

Laser Threshold

A laser below threshold ($r < 0$) has zero output (stable $x^* = 0$). Above threshold ($r > 0$), $x^* = 0$ becomes unstable and a new stable state $x^* = r$ (proportional to the laser output intensity) takes over. This is a transcritical bifurcation.

2.4 Pitchfork Bifurcation

Pitchfork bifurcations arise in systems that possess a **symmetry** $x \to -x$. There are two types.

2.4.1 Supercritical Pitchfork

Normal Form: Supercritical Pitchfork

$$\dot{x} = rx - x^3. \tag{9}$$

Analysis. Fixed points: $x^* = 0$ for all r, and $x^* = \pm\sqrt{r}$ for $r > 0$.

Since $f'(x) = r - 3x^2$:

- $f'(0) = r$: origin is stable for $r < 0$, unstable for $r > 0$.
- $f'(\pm\sqrt{r}) = r - 3r = -2r < 0$ for $r > 0$: the two new branches are stable.

At $r = 0$, the origin loses stability and two new stable branches are born symmetrically. The bifurcation diagram looks like a "pitchfork."

2.4.2 Subcritical Pitchfork

Normal Form: Subcritical Pitchfork

$$\dot{x} = rx + x^3. \tag{10}$$

Here the cubic term is *destabilizing*. For $r < 0$, there are three fixed points: the origin (stable) and $x^* = \pm\sqrt{-r}$ (both unstable). At $r = 0$ the unstable branches merge with the origin, making it unstable for $r > 0$.

Subcritical Pitchfork and Hysteresis

In a physical system, the subcritical pitchfork is often "stabilized" by a higher-order term, e.g., $\dot{x} = rx + x^3 - x^5$. This leads to **hysteresis**: the system can jump discontinuously between branches as r is varied, and the value of x observed depends on the *direction* in which r is changed.

2.5 Imperfect Bifurcations and Catastrophes

When the symmetry of a pitchfork bifurcation is slightly broken (an "imperfection" h), the bifurcation diagram deforms. The normal form is

$$\dot{x} = h + rx - x^3. \tag{11}$$

For $h \neq 0$, the perfect pitchfork breaks into an isolated branch and an S-shaped curve. The set of bifurcation points in the (r, h) parameter plane forms a cusp—this is the **cusp catastrophe**, one of the elementary catastrophes of singularity theory.

2.6 Summary of 1D Bifurcations

Bifurcation	**Normal Form**	**Mechanism**
Saddle-node	$\dot{x} = r + x^2$	Two FPs created/destroyed
Transcritical	$\dot{x} = rx - x^2$	FPs exchange stability
Supercrit. pitchfork	$\dot{x} = rx - x^3$	One FP $\to$ three; stable branches
Subcrit. pitchfork	$\dot{x} = rx + x^3$	Three FPs $\to$ one; unstable branches

2.7 Application: Overdamped Bead on a Rotating Hoop

A bead slides without friction on a circular wire hoop of radius R that rotates about its vertical diameter with angular velocity ω. In the overdamped limit (viscous damping dominates inertia), the angle ϕ of the bead from the bottom of the hoop obeys:

$$\dot{\phi} = f(\phi) = \sin\phi\big(\gamma\cos\phi - 1\big), \tag{12}$$

where $\gamma = R\omega^2/g$ is the dimensionless rotation rate.

Analysis:

- Fixed points: $\phi^* = 0, \pi$ always exist. For $\gamma > 1$, two additional fixed points appear at $\phi^* = \pm\arccos(1/\gamma)$.
- At $\gamma = 1$: a **supercritical pitchfork bifurcation** occurs. The bottom ($\phi = 0$) changes from stable to unstable, and the bead spontaneously climbs to one of the new off-axis equilibria.

This is a beautiful physical realization of symmetry-breaking: for slow rotation the bead sits at the bottom; for fast rotation it "chooses" a side, breaking the left-right symmetry.

2.8 Application: Insect Outbreak (Spruce Budworm)

The budworm population N satisfies a model of the form

$$\dot{N} = rN\left(1 - \frac{N}{K}\right) - p(N), \tag{13}$$

where the first term is logistic growth and $p(N) = BN^2/(A^2 + N^2)$ is predation (a sigmoidal functional response). Nondimensionalizing with $x = N/A$, $\tau = rAt$, etc., leads to

$$\dot{x} = rx\left(1 - \frac{x}{k}\right) - \frac{x^2}{1 + x^2}. \tag{14}$$

For appropriate parameter ranges, this system exhibits **hysteresis** via saddle-node bifurcations. As parameters change slowly (e.g., the forest matures, increasing k), the budworm population can undergo a sudden catastrophic outbreak (a jump to a high-x branch), followed by an equally sudden crash when the branch disappears. This involves two saddle-node bifurcations forming a hysteresis loop—an example of a **cusp catastrophe** in an ecological setting.

3 Flows on the Circle

3.1 Motivation and Definitions

On the real line, oscillations are impossible for a first-order autonomous system. However, if the phase space is a **circle** S^1 (i.e., the variable θ is 2π-periodic), oscillations *are* possible: the system can flow around the circle indefinitely.

Definition 3.1. A **flow on the circle** is an ODE of the form

$$\dot{\theta} = f(\theta), \qquad \theta \in [0, 2\pi), \tag{15}$$

where $f(\theta + 2\pi) = f(\theta)$.

Fixed points are again defined by $f(\theta^*) = 0$, and stability analysis is identical to the real-line case.

3.2 Uniform and Nonuniform Oscillators

Uniform oscillator: $\dot{\theta} = \omega$ (constant angular velocity). Period $T = 2\pi/\omega$.

Nonuniform oscillator: $\dot{\theta} = \omega - a\sin\theta$ with $a, \omega > 0$.

- If $\omega > a$: $\dot{\theta} > 0$ always, so θ oscillates but nonuniformly. The period is

$$T = \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \tag{16}$$

- At $\omega = a$: a saddle-node bifurcation creates a half-stable fixed point. The period diverges: $T \to \infty$.
- For $\omega < a$: two fixed points appear (one stable, one unstable) and oscillation ceases.

This is the prototypical example of a **saddle-node bifurcation on a circle**, also called a SNIPER (Saddle-Node Infinite-PERiod) bifurcation. The divergence of the period scales as $T \sim 1/\sqrt{\omega - a}$ near the bifurcation.

3.3 The Overdamped Pendulum

A pendulum in a viscous fluid with a constant applied torque Γ is modeled by

$$\dot{\theta} = \Gamma - \sin\theta. \tag{17}$$

This is exactly the nonuniform oscillator with $\omega = \Gamma$ and $a = 1$. For $|\Gamma| < 1$, the pendulum settles to a tilted equilibrium; for $|\Gamma| > 1$, it whirls continuously.

3.4 Superconducting Josephson Junctions

A Josephson junction consists of two superconductors separated by a thin insulating barrier. The *phase difference* ϕ across the junction obeys (in the overdamped, zero-capacitance limit):

$$\frac{\hbar}{2eR}\,\dot{\phi} = I - I_c \sin\phi, \tag{18}$$

where I is the applied (DC) current, I_c is the critical current, and R is the shunt resistance. Nondimensionalizing:

$$\dot{\phi} = i - \sin\phi, \qquad i \equiv I/I_c. \tag{19}$$

This is a flow on the circle identical to the overdamped pendulum.

- For $|i| < 1$: two fixed points exist; $\dot{\phi} = 0$ at steady state and the voltage $V = (\hbar/2e)\,\dot{\phi} = 0$. The junction carries a **supercurrent** with zero resistance.
- For $|i| > 1$: no fixed points; ϕ winds continuously and a nonzero DC voltage appears across the junction.

Part II
Two-Dimensional Flows

4 Linear Systems in Two Dimensions

4.1 Setup

We now consider systems of two coupled first-order ODEs:

$$\dot{\vec{x}} = A\vec{x}, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}. \tag{20}$$

4.2 Solution by Eigenvalues and Eigenvectors

We seek solutions of the form $\vec{x}(t) = e^{\lambda t}\vec{v}$. Substituting into (20):

$$\lambda e^{\lambda t}\vec{v} = A\,e^{\lambda t}\vec{v} \quad \Longrightarrow \quad A\vec{v} = \lambda\vec{v}.$$

Thus λ must be an **eigenvalue** of A and $\vec{v}$ the corresponding **eigenvector**.

The eigenvalues satisfy the characteristic equation:

$$\lambda^2 - \tau\lambda + \Delta = 0, \tag{21}$$

where $\tau \equiv \operatorname{tr} A = a + d$ and $\Delta \equiv \det A = ad - bc$. The solutions are:

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}. \tag{22}$$

The general solution is $\vec{x}(t) = c_1 e^{\lambda_1 t}\vec{v}_1 + c_2 e^{\lambda_2 t}\vec{v}_2$ (when the eigenvalues are distinct).

4.3 Classification of Fixed Points

The qualitative behavior of trajectories near the origin depends on the eigenvalues, which are determined by τ and Δ.

Classification in the (τ, Δ) Plane

Region	Eigenvalues	Fixed Point Type
$\Delta < 0$	Real, opposite signs	**Saddle point**
$\Delta > 0$, $\tau^2 > 4\Delta$, $\tau < 0$	Real, both negative	**Stable node**
$\Delta > 0$, $\tau^2 > 4\Delta$, $\tau > 0$	Real, both positive	**Unstable node**
$\Delta > 0$, $\tau^2 < 4\Delta$, $\tau < 0$	Complex, Re < 0	**Stable spiral**
$\Delta > 0$, $\tau^2 < 4\Delta$, $\tau > 0$	Complex, Re > 0	**Unstable spiral**
$\Delta > 0$, $\tau = 0$	Pure imaginary	**Center** (linear)
$\Delta > 0$, $\tau^2 = 4\Delta$	Real, repeated	**Star node** or **degenerate node**
$\Delta = 0$	One eigenvalue zero	**Line of fixed points**

Key relationships:

$$\lambda_1 + \lambda_2 = \tau = \operatorname{tr} A, \tag{23}$$

$$\lambda_1 \cdot \lambda_2 = \Delta = \det A. \tag{24}$$

The stability boundary is $\tau = 0$ (eigenvalues cross the imaginary axis). The boundary between nodes and spirals is $\tau^2 = 4\Delta$. Saddle points live below the $\Delta = 0$ axis.

Physical Interpretation

- **Stable node:** trajectories approach the origin along straight lines (the eigenvector directions); overdamped approach.
- **Stable spiral:** trajectories spiral inward; underdamped oscillatory decay.
- **Center:** closed elliptical orbits; no damping, no growth.
- **Saddle:** trajectories approach along one eigenvector direction and are repelled along the other.

5 Phase Plane Analysis

5.1 Nonlinear Systems and Phase Portraits

We now consider the general two-dimensional nonlinear system:

$$\dot{x} = f(x, y), \qquad \dot{y} = g(x, y). \tag{25}$$

The **phase portrait** is the collection of all trajectories in the (x, y) plane.

Definition 5.1 (Nullclines)**.** The **x-nullcline** is the set of points where $\dot{x} = 0$, i.e., $f(x, y) = 0$. The **y-nullcline** is the set where $\dot{y} = 0$, i.e., $g(x, y) = 0$. Fixed points lie at the *intersections* of the two nullclines.

Nullclines are extremely useful for sketching phase portraits: on the x-nullcline, the flow is purely vertical ($\dot{x} = 0$); on the y-nullcline, the flow is purely horizontal ($\dot{y} = 0$).

5.2 Existence, Uniqueness, and Topological Consequences

Theorem 5.1 (Existence and Uniqueness in 2D)**.** *If f and g are continuously differentiable, then given any initial condition (x_0, y_0), a unique solution exists (at least locally in time).*

Corollary 5.2. *Different trajectories can never cross in the phase plane.*

This has profound consequences: for example, a closed orbit divides the plane into an "inside" and an "outside," and no trajectory can cross it.

5.3 Linearization Near Fixed Points

Let (x^*, y^*) be a fixed point of (25). Set $u = x - x^*$, $v = y - y^*$. Taylor expanding:

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = J\Big|_{(x^*,y^*)} \begin{pmatrix} u \\ v \end{pmatrix} + \text{h.o.t.}, \tag{26}$$

where J is the **Jacobian matrix**:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix}\Bigg|_{(x^*,y^*)}. \tag{27}$$

Theorem 5.3 (Hartman–Grobman Theorem—Informal)**.** *If the fixed point is* ***hyperbolic*** *(i.e., $Re(\lambda_i) \neq 0$ for all eigenvalues of J), then the phase portrait of the nonlinear system near the fixed point is* qualitatively the same *as the phase portrait of the linearized system $\dot{\vec{u}} = J\vec{u}$.*

Borderline Cases

- **Centers** ($\tau = 0$, $\Delta > 0$) are *not* hyperbolic. A linear center may become a spiral (stable or unstable) in the nonlinear system. Additional analysis (e.g., conserved quantities, reversibility) is needed.
- **Non-isolated fixed points** ($\Delta = 0$) also require special treatment.

5.4 Application: Competing Species (Rabbits vs. Sheep)

Consider the Lotka–Volterra competition model:

$$\dot{x} = x(3 - x - 2y), \qquad \dot{y} = y(2 - x - y), \tag{28}$$

where x and y represent two competing populations.

Nullclines: $\dot{x} = 0$: $x = 0$ or $x + 2y = 3$. $\dot{y} = 0$: $y = 0$ or $x + y = 2$.

Fixed points: $(0, 0)$, $(3, 0)$, $(0, 2)$, and the intersection of $x + 2y = 3$ and $x + y = 2$, giving $(1, 1)$.

Linearizing at each fixed point using the Jacobian:

- $(0, 0)$: $J = \left(\begin{smallmatrix} 3 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, eigenvalues $3, 2 > 0$: unstable node.
- $(3, 0)$: $J = \left(\begin{smallmatrix} -3 & -6 \\ 0 & -1 \end{smallmatrix}\right)$, eigenvalues $-3, -1$: stable node (sheep extinct).
- $(0, 2)$: $J = \left(\begin{smallmatrix} -1 & 0 \\ -2 & -2 \end{smallmatrix}\right)$, eigenvalues $-1, -2$: stable node (rabbits extinct).
- $(1, 1)$: $\tau = -3$, $\Delta = 1$, $\tau^2 - 4\Delta = 5 > 0$: a saddle point (coexistence equilibrium is unstable).

The separatrix of the saddle at $(1, 1)$ divides the first quadrant into two basins of attraction: one leading to $(3, 0)$ (species x wins) and the other to $(0, 2)$ (species y wins). The outcome depends on initial conditions—**competitive exclusion**.

5.5 Conservative Systems

Definition 5.2 (Conservative System)**.** A system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ is **conservative** if there exists a function $E(x, y)$ (the *energy* or *conserved quantity*) such that E is constant along trajectories: $\frac{\mathrm{d}E}{\mathrm{d}t} = 0$.

Theorem 5.4. *Conservative systems have* ***no attractors****. Trajectories lie on level sets of E, and there can be no stable nodes, stable spirals, or stable limit cycles.*

Proof. Since E is constant along trajectories, no trajectory can approach a single point (which would require E to take a single value over an open neighborhood), nor can a trajectory spiral inward (which would require E to decrease monotonically). □

Example 5.1 (Harmonic Oscillator)**.** $m\ddot{x} = -kx$ can be written as $\dot{x} = v$, $\dot{v} = -(k/m)\,x$. The energy $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$ is conserved:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = kx\dot{x} + mv\dot{v} = kxv + mv\big(-\tfrac{k}{m}x\big) = 0.$$

The Jacobian at the origin has $\tau = 0$, $\Delta = k/m > 0$: a linear center. Since the system is conservative, this center persists as a true nonlinear center with closed elliptical orbits.

5.6 Reversible Systems

Definition 5.3 (Time-Reversibility)**.** A system is **reversible** if it is invariant under the transformation $t \to -t$ and some transformation of the state variables (typically $y \to -y$, or equivalently, the velocity reverses).

Theorem 5.5. *For a reversible system, if a trajectory lies on the line of symmetry (the axis that is invariant under the reversal), then any linear center is a* true *nonlinear center.*

Example 5.2 (Undamped Pendulum)**.** The equation $\ddot{\theta} = -\sin\theta$ written as a system: $\dot{\theta} = v$, $\dot{v} = -\sin\theta$. Under $t \to -t$, $v \to -v$: the equations are unchanged. Therefore the linear center at $(\theta, v) = (0, 0)$ is a true nonlinear center.

The energy is $E = \frac{1}{2}v^2 - \cos\theta$. Phase portrait:

- $E < 1$: librations (closed orbits around the center).
- $E = 1$: the **separatrix** (homoclinic orbits connecting the saddle at $(\pi, 0)$ to itself).
- $E > 1$: rotations (unbounded trajectories wrapping around).

5.7 Index Theory

Index theory provides topological constraints on the arrangement of fixed points and closed orbits.

Definition 5.4 (Index of a Closed Curve)**.** Let C be a simple closed curve in the phase plane that does not pass through any fixed points. At each point on C, the vector field $(\dot{x}, \dot{y})$ makes an angle ϕ with the positive x-axis. As we traverse C counterclockwise, ϕ changes continuously. The **index** of C is

$$I_C = \frac{1}{2\pi}\oint_C d\phi = \frac{[\phi]_C}{2\pi}, \tag{29}$$

i.e., the number of complete counterclockwise revolutions of the vector field as we traverse C once counterclockwise.

Theorem 5.6 (Properties of the Index). *1. If C encloses no fixed points, then $I_C = 0$.*

2. *If C is a trajectory (e.g., a closed orbit), then $I_C = +1$.*
3. *The index is unchanged if C is continuously deformed, as long as it does not cross a fixed point.*
4. *The index of a curve enclosing multiple isolated fixed points equals the sum of the indices of the individual fixed points.*
5. *Reversing time ($t \to -t$) does not change the index.*

Indices of Common Fixed Points

Fixed Point Type	Index
Stable/unstable node	$+1$
Stable/unstable spiral	$+1$
Center	$+1$
Saddle point	-1

Corollary 5.7. *Any closed orbit must enclose fixed points whose indices sum to $+1$. In particular, a closed orbit must enclose at least one fixed point that is not a saddle.*

6 Limit Cycles

6.1 Definitions and Examples

Definition 6.1 (Limit Cycle). A **limit cycle** is an isolated closed orbit. "Isolated" means that neighboring trajectories are not closed—they either spiral toward or away from the limit cycle.

- **Stable** limit cycle: all nearby trajectories approach it as $t \to \infty$.
- **Unstable** limit cycle: all nearby trajectories move away.
- **Half-stable**: trajectories approach from one side and recede from the other.

Remark 6.1. Limit cycles are inherently **nonlinear** phenomena. They cannot occur in linear systems (where closed orbits, if they exist, form continuous families—centers—not isolated cycles).

Example 6.1 (Limit Cycle in Polar Coordinates). Consider $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. The radial equation has fixed points $r^* = 0$ (unstable, since $f'(0) = 1 > 0$) and $r^* = 1$ (stable, since $f'(1) = -2 < 0$). Therefore $r = 1$ is a **stable limit cycle**: all trajectories with $r > 0$ spiral toward the unit circle.

6.2 Ruling Out Closed Orbits

Before searching for limit cycles, it is often useful to rule them out in certain regions.

6.2.1 Gradient Systems

If $\dot{x} = -\nabla V$ for some potential $V(x, y)$, then

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \nabla V \cdot \dot{\vec{x}} = \nabla V \cdot (-\nabla V) = -|\nabla V|^2 \leq 0.$$

So V decreases monotonically along trajectories, and closed orbits are impossible.

6.2.2 Liapunov Functions

Theorem 6.1 (Liapunov's Theorem for Ruling Out Closed Orbits)**.** *If there exists a function $V(x, y)$ satisfying:*

1. *$V(x^*, y^*) = 0$ at all fixed points,*
2. *$V(x, y) > 0$ for all (x, y) not a fixed point (in some region R),*
3. *$\dot{V} = \frac{\mathrm{d}V}{\mathrm{d}t} < 0$ for all non-fixed points in R,*

then there are no closed orbits entirely contained in R.

Proof sketch. On a closed orbit, V would have to return to its starting value after one period. But $\dot{V} < 0$ means V is strictly decreasing, a contradiction. □

6.2.3 Dulac's Criterion

Theorem 6.2 (Dulac's Criterion)**.** *Let R be a simply connected region of the plane. If there exists a continuously differentiable function $g(x, y)$ such that $\nabla \cdot (g\vec{F})$ does not change sign in R (and is not identically zero on any open subset), then there are no closed orbits entirely contained in R.*

Here $\vec{F} = (f, g_{sys})$ is the vector field, and we require

$$\frac{\partial(g\, f)}{\partial x} + \frac{\partial(g\, g_{sys})}{\partial y} \neq 0 \quad \textit{(same sign throughout } R\textit{).} \tag{30}$$

Proof. By Green's theorem, for any closed curve C bounding a region $A \subset R$:

$$\iint_A \nabla \cdot (g\vec{F})\, dA = \oint_C g\vec{F} \cdot \hat{n}\, dl.$$

If C were a closed orbit, then $\vec{F}$ is tangent to C everywhere, so $\vec{F} \cdot \hat{n} = 0$ and the right side vanishes. But the left side is nonzero by hypothesis, a contradiction. □

6.3 Poincaré–Bendixson Theorem

This theorem is one of the most powerful tools for *proving the existence* of limit cycles.

Theorem 6.3 (Poincaré–Bendixson Theorem)**.** *Suppose that:*

1. *R is a closed, bounded region of the plane;*
2. *$\dot{\vec{x}} = \vec{F}(\vec{x})$ is a continuously differentiable vector field on an open set containing R;*
3. *R does not contain any fixed points;*

4. *There exists a trajectory C that is "confined" to R (i.e., C starts in R and can never leave).*

Then C is either a closed orbit, or it spirals toward a closed orbit contained in R. In either case, R contains a ***closed orbit****.*

Remark 6.2. This theorem is specific to two dimensions. In three or more dimensions, a bounded trajectory need not approach a closed orbit (it could be a strange attractor, for example).

Trapping Regions

To apply the Poincaré–Bendixson theorem, one constructs a **trapping region**: a closed, bounded region R with no fixed points inside, such that the vector field points inward (or is tangent) on the entire boundary ∂R. Then any trajectory entering R is confined to it, and the theorem guarantees a closed orbit inside.
The boundary of R is often built from segments of nullclines, coordinate axes, or level curves of auxiliary functions.

6.4 Liénard Systems

Definition 6.2 (Liénard System)**.** A **Liénard system** is a second-order ODE of the form

$$\ddot{x} + f(x)\dot{x} + g(x) = 0, \tag{31}$$

or equivalently, the system $\dot{x} = y - F(x)$, $\dot{y} = -g(x)$, where $F(x) = \int_0^x f(s)\,ds$.

Theorem 6.4 (Liénard's Theorem)**.** *The Liénard system has a unique, stable limit cycle if:*

1. *f and g are continuous, and g is Lipschitz;*
2. *g is an odd function: $g(-x) = -g(x)$, with $g(x) > 0$ for $x > 0$;*
3. *F is an odd function with exactly one positive zero at $x = a$, $F(x) < 0$ for $0 < x < a$, $F(x) > 0$ and nondecreasing for $x > a$, and $F(x) \to \infty$ as $x \to \infty$.*

6.5 The Van der Pol Oscillator

The most important Liénard system is the **Van der Pol oscillator**:

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \qquad \mu > 0. \tag{32}$$

Here $f(x) = \mu(x^2 - 1)$ and $g(x) = x$. The damping is *negative* for $|x| < 1$ (energy is pumped in) and *positive* for $|x| > 1$ (energy is dissipated). This self-regulating mechanism produces a unique, stable limit cycle for any $\mu > 0$.

Rewriting as a system with $y = \dot{x}/\mu + x^3/3 - x$ (Liénard plane):

$$\dot{x} = \mu\big(y - F(x)\big), \qquad \dot{y} = -\frac{x}{\mu}, \qquad F(x) = \frac{x^3}{3} - x. \tag{33}$$

6.6 Relaxation Oscillations

For $\mu \gg 1$ (strong nonlinearity), the Van der Pol oscillator exhibits **relaxation oscillations**: the motion consists of slow crawls along the cubic nullcline $y = F(x) = x^3/3 - x$ punctuated by fast jumps between the two branches.

Mechanism:

1. **Fast dynamics** ($\dot{x}$ is large): the trajectory jumps nearly horizontally between the left and right branches of the S-shaped nullcline.
2. **Slow dynamics** ($\dot{y}$ is small): the trajectory crawls along the nullcline until it reaches a knee (fold), then jumps again.

The period of relaxation oscillations scales as $T \approx (3 - 2\ln 2)\mu \approx 1.614\,\mu$ for large μ—it grows linearly with the nonlinearity parameter, in contrast to the nearly constant period of weakly nonlinear oscillations.

7 Bifurcations Revisited: Two-Dimensional Systems

7.1 Saddle-Node, Transcritical, and Pitchfork in 2D

In two dimensions, all three bifurcation types from Section 2 can occur. The mechanism is the same: one eigenvalue of the Jacobian passes through zero. The other eigenvalue remains nonzero, so the dynamics in the "transverse" direction is simply attracted to (or repelled from) a manifold on which the 1D bifurcation takes place.

7.2 Hopf Bifurcation

The **Hopf bifurcation** is the birth (or death) of a limit cycle from a fixed point. It occurs when a pair of complex conjugate eigenvalues crosses the imaginary axis.

Theorem 7.1 (Hopf Bifurcation Theorem—Informal)**.** *Consider $\dot{\vec{x}} = \vec{F}(\vec{x};\mu)$ with a fixed point at the origin. Suppose the Jacobian at the origin has eigenvalues $\lambda(\mu) = \alpha(\mu) \pm i\omega(\mu)$. If:*

1. *At $\mu = 0$: $\alpha(0) = 0$ and $\omega(0) \neq 0$ (pure imaginary eigenvalues);*
2. *$\alpha'(0) \neq 0$ (eigenvalues cross the imaginary axis with nonzero speed);*

then a Hopf bifurcation occurs at $\mu = 0$.

7.2.1 Supercritical Hopf Bifurcation

A **stable limit cycle** is born from the fixed point as it loses stability.

Normal Form: Supercritical Hopf

In polar coordinates:

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega + br^2. \tag{34}$$

- For $\mu < 0$: origin is a stable spiral.
- For $\mu > 0$: origin becomes unstable; a stable limit cycle appears at $r = \sqrt{\mu}$.

The amplitude of the limit cycle grows as $\sqrt{\mu}$, and its frequency is $\omega + b\mu$ (the parameter b shifts the frequency, while ω sets the base frequency).

7.2.2 Subcritical Hopf Bifurcation

An **unstable** limit cycle shrinks down to the origin and is absorbed as the origin gains stability.

Normal form in polar coordinates:

$$\dot{r} = \mu r + r^3, \qquad \dot{\theta} = \omega + br^2. \tag{35}$$

- For $\mu < 0$: an unstable limit cycle at $r = \sqrt{-\mu}$ coexists with a stable origin.
- At $\mu = 0$: the limit cycle collapses into the origin.
- For $\mu > 0$: the origin is unstable and no small-amplitude limit cycle exists. Trajectories may jump to a distant attractor, leading to a **hard** or **catastrophic** transition.

7.3 Global Bifurcations of Cycles

Limit cycles can also be born or destroyed through *global* bifurcations that involve large-scale rearrangements of the phase portrait.

1. **Saddle-node bifurcation of cycles (fold of cycles):** A stable and an unstable limit cycle collide and annihilate.
2. **Homoclinic bifurcation:** A limit cycle grows until it touches a saddle point, forming a *homoclinic orbit* (a trajectory that connects the saddle to itself). As the parameter passes through the bifurcation, the cycle's period diverges ($T \to \infty$), and the cycle disappears.
3. **Infinite-period (SNIPER) bifurcation:** A limit cycle passes through a saddle-node on a circle. The period diverges but the amplitude remains finite.

7.4 Hysteresis in the Josephson Junction (2D)

With capacitance included, the Josephson junction becomes a second-order system:

$$i = \sin\phi + \alpha\dot{\phi} + \ddot{\phi}, \tag{36}$$

which is equivalent to the **damped driven pendulum**:

$$\dot{\phi} = y, \qquad \dot{y} = i - \sin\phi - \alpha y. \tag{37}$$

The Jacobian is $J = \left(\begin{smallmatrix} 0 & 1 \\ -\cos\phi & -\alpha \end{smallmatrix}\right)$.

The parameter α controls the damping:

- **Overdamped** ($\alpha \gg 1$): reduces to the 1D model of Section 3.4.
- **Underdamped** ($\alpha < 1$): hysteresis occurs. As i increases past 1, the zero-voltage state is lost; but when i is decreased, the system remains on the rotating (voltage) branch until i drops below a retrapping current $i_r < 1$.

7.5 Coupled Oscillators and Quasiperiodicity

Consider two oscillators with phase variables θ_1, θ_2 on the circle, coupled weakly:

$$\dot{\theta}_1 = \omega_1 + K_1 \sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = \omega_2 + K_2 \sin(\theta_1 - \theta_2). \tag{38}$$

Defining the phase difference $\phi = \theta_1 - \theta_2$:

$$\dot{\phi} = (\omega_1 - \omega_2) - (K_1 + K_2) \sin \phi. \tag{39}$$

This is again a flow on the circle!

- **Phase locking:** If $|\omega_1 - \omega_2| < K_1 + K_2$, stable fixed points exist: the oscillators lock to a constant phase difference.
- **Phase drift:** If $|\omega_1 - \omega_2| > K_1 + K_2$, no fixed points; ϕ drifts and the oscillators are **not synchronized**.

7.5.1 Dynamics on the Torus

The combined phase space (θ_1, θ_2) lives on a **torus** $\mathbb{T}^2 = S^1 \times S^1$. When uncoupled ($K_1 = K_2 = 0$), each oscillator moves independently, and the trajectory on the torus is either:

- A **closed orbit** if ω_1/ω_2 is rational, or
- A **quasiperiodic orbit** (densely filling the torus) if ω_1/ω_2 is irrational.

7.6 Poincaré Maps

Definition 7.1 (Poincaré Map)**.** Choose a **section** Σ (a codimension-1 surface transverse to the flow) in the phase space. The **Poincaré map** (or first-return map) $P : \Sigma \to \Sigma$ sends each point $\vec{x} \in \Sigma$ to the next point where the trajectory through $\vec{x}$ intersects Σ.

This reduces a continuous-time system to a discrete-time map, lowering the dimension by one. A **closed orbit** of the flow corresponds to a **fixed point** of the Poincaré map.

Part III
Chaos

8 The Lorenz Equations

8.1 Origin and Physical Motivation

Edward Lorenz (1963) derived a simplified model of atmospheric convection (Rayleigh–Bénard convection). The system consists of three coupled ODEs:

$$\boxed{\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - y - xz, \qquad \dot{z} = xy - bz,} \tag{40}$$

where $\sigma > 0$ is the Prandtl number, $r > 0$ is (proportional to) the Rayleigh number, and $b > 0$ is a geometric factor. All parameters are positive. The classic parameter values are $\sigma = 10$, $b = 8/3$.

8.2 Simple Properties

1. **Symmetry:** The system is invariant under $(x, y, z) \to (-x, -y, z)$.

2. **Dissipation:** The divergence of the flow is

$$\nabla \cdot \vec{F} = -\sigma - 1 - b < 0,$$

so volumes in phase space contract exponentially. This means the system is **dissipative** and all trajectories eventually become confined to an attractor of zero volume.

3. **Fixed points:**

   - The origin $O = (0, 0, 0)$ exists for all r.
   - For $r > 1$, two additional fixed points appear via a **supercritical pitchfork** bifurcation:

$$C^{\pm} = \big(\pm\sqrt{b(r-1)},\ \pm\sqrt{b(r-1)},\ r-1\big).$$

4. **Stability of the origin:** The eigenvalues at O are found from the Jacobian. The origin is stable for $r < 1$ and unstable (a saddle with one unstable direction) for $r > 1$.

5. **Stability of $C^{\pm}$:** These are stable spirals for $1 < r < r_H$, where

$$r_H = \sigma\,\frac{\sigma + b + 3}{\sigma - b - 1} \tag{41}$$

(assuming $\sigma > b + 1$). At $r = r_H$, a **subcritical Hopf bifurcation** occurs: $C^{\pm}$ lose stability and an unstable limit cycle is absorbed.

6. **Trapping region:** Using the Liapunov-like function $V = rx^2 + \sigma y^2 + \sigma(z - 2r)^2$, one can show $\dot{V} < 0$ outside a large ellipsoid. Hence all trajectories eventually enter this region and remain bounded.

8.3 Chaos on the Strange Attractor

For $r > r_H$ (and the classic parameters), trajectories exhibit **chaotic** behavior:

- **Aperiodic:** Trajectories never repeat, even though they are bounded.
- **Sensitive dependence on initial conditions:** Nearby trajectories diverge exponentially. This is the hallmark of chaos.
- **Strange attractor:** The attractor has a fractal structure with zero volume but infinite detail.

Definition 8.1 (Sensitive Dependence and Horizon Time)**.** Two nearby initial conditions separated by δ_0 diverge roughly as $\delta(t) \sim \delta_0\, e^{\lambda t}$, where $\lambda > 0$ is the **Lyapunov exponent**. Predictions become unreliable when $\delta(t) \sim \epsilon$ (some tolerance). The **horizon time** is:

$$t_h \sim \frac{1}{\lambda} \ln \frac{\epsilon}{\delta_0}. \tag{42}$$

Beyond t_h, prediction is essentially impossible despite deterministic dynamics. Note the logarithmic dependence: even dramatic improvements in measurement precision δ_0 yield only modest increases in t_h.

8.4 The Lorenz Map

Lorenz discovered that the chaotic dynamics can be partially understood by a one-dimensional map. Record the successive local maxima z_n of $z(t)$. Plotting z_{n+1} vs. z_n yields a remarkably thin, single-humped curve (approximately a tent map).

Key observations:

- The slope $|f'| > 1$ everywhere on the Lorenz map, implying that nearby orbits are stretched apart at each return.
- There are no stable fixed points or periodic orbits of the map, explaining the aperiodic behavior.
- The thinness of the map (nearly 1D) arises from the extreme contraction in the dissipative system.

8.5 Exploring Parameter Space of the Lorenz System

As r increases (with $\sigma = 10$, $b = 8/3$), the Lorenz system passes through several regimes:

1. $0 < r < 1$: The origin is the global attractor. All motion decays.
2. $1 < r < r_H \approx 24.74$: The fixed points $C^{\pm}$ are stable spirals. Transient chaos may occur for $r > 13.9$ (where a "strange saddle" exists), but all trajectories eventually settle to C^+ or C^-.
3. $r_H < r$: Chaotic attractor (the "butterfly"). The system exhibits sustained aperiodic motion, sensitive dependence, and a fractal attractor.
4. Periodic windows exist within the chaotic regime, where stable periodic orbits briefly emerge.

8.6 The AC-Driven Pendulum and Routes to Chaos

The damped pendulum with a periodic (AC) driving torque provides another canonical route to chaos:

$$\Gamma \cos(\omega t) = \ddot{\theta} + b\dot{\theta} + \sin\theta. \tag{43}$$

Writing as a system with $\dot{\theta} = v$, $\dot{v} = -bv - \sin\theta + \Gamma\cos(\omega y)$, $\dot{y} = 1$, this is a three-dimensional autonomous system (the explicit time dependence elevates it to 3D).

The Poincaré section (strobing at the drive frequency) reduces it to a 2D map. As Γ increases, the system passes through:

- Periodic (phase-locked) motion,
- Period-doubling cascades,
- Chaotic tumbling,

providing a concrete physical system exhibiting the period-doubling route to chaos.

9 What Is Chaos?

Before proceeding to maps, let us formalize the notion of chaos.

Definition 9.1 (Chaos—Working Definition)**.** A dynamical system is **chaotic** if it satisfies all three conditions:

1. **Aperiodic long-term behavior:** Trajectories do not settle to fixed points, periodic orbits, or quasiperiodic orbits.
2. **Deterministic:** There is no randomness in the equations—the system is governed by definite rules with no stochastic terms.
3. **Sensitive dependence on initial conditions:** Nearby trajectories diverge exponentially, with at least one positive Lyapunov exponent.

Remark 9.1. Chaos is *not* randomness. The underlying equations are perfectly deterministic; the apparent randomness arises from the amplification of microscopic uncertainties in initial conditions. This is sometimes called **deterministic chaos** or **deterministic aperiodicity**.

Remark 9.2. Chaos requires at least three dimensions for continuous-time (flow) systems. In two dimensions, the Poincaré–Bendixson theorem constrains bounded trajectories to approach fixed points or closed orbits. However, for discrete-time systems (maps), chaos is possible even in one dimension (e.g., the logistic map).

10 One-Dimensional Maps

10.1 Introduction

Motivated by Poincaré maps and the Lorenz map, we now study discrete-time dynamical systems (**iterated maps**) of the form

$$x_{n+1} = f(x_n), \tag{44}$$

where $f : \mathbb{R} \to \mathbb{R}$ is a given function.

10.2 Fixed Points and Cobweb Diagrams

Definition 10.1. A point x^* is a **fixed point** of the map (44) if $f(x^*) = x^*$.

Graphically, fixed points are found by intersecting the graph of $f(x)$ with the diagonal line $y = x$.

Stability of a Fixed Point (Maps)

Let x^* be a fixed point of $x_{n+1} = f(x_n)$. The **multiplier** is $\mu = f'(x^*)$.

- $|\mu| < 1$: x^* is **stable** (attracting).
- $|\mu| > 1$: x^* is **unstable** (repelling).
- $|\mu| = 1$: inconclusive.

Proof. Let $\delta_n = x_n - x^*$. Then $\delta_{n+1} = f(x^* + \delta_n) - x^* \approx f'(x^*)\,\delta_n$, so $\delta_n \approx [f'(x^*)]^n \delta_0$. □

Cobweb construction: A graphical method for iterating maps. Starting from $(x_0, 0)$:

1. Go vertically to $y = f(x_0)$.
2. Go horizontally to the diagonal $y = x$ (so the horizontal coordinate is now $x_1 = f(x_0)$).
3. Repeat.

10.3 Periodic Orbits

Definition 10.2. A point p has **period** n if $f^n(p) = p$ and $f^k(p) \neq p$ for $0 < k < n$, where f^n denotes the n-fold composition $f \circ f \circ \cdots \circ f$.

A period-n orbit is a fixed point of f^n. Its stability is determined by the multiplier of f^n:

$$(f^n)'(p) = f'(p) \cdot f'(f(p)) \cdot f'(f^2(p)) \cdots f'(f^{n-1}(p)) = \prod_{k=0}^{n-1} f'(f^k(p)). \tag{45}$$

The orbit is stable if $|(f^n)'(p)| < 1$ and unstable if $|(f^n)'(p)| > 1$.

10.4 The Logistic Map

The most studied one-dimensional map is the **logistic map**:

$$x_{n+1} = rx_n(1 - x_n), \qquad 0 \leq r \leq 4. \tag{46}$$

10.4.1 Fixed Points

Setting $x = rx(1 - x)$: the fixed points are $x^* = 0$ and $x^* = (r - 1)/r$ (for $r > 1$).

Stability of $x^* = (r - 1)/r$: $f'(x) = r - 2rx$, so $f'(x^*) = r - 2(r - 1) = 2 - r$. This fixed point is stable when $|2 - r| < 1$, i.e., $1 < r < 3$.

10.4.2 Period Doubling

At $r = 3$, the fixed point $x^* = (r - 1)/r$ loses stability through a **flip bifurcation** ($f'(x^*) = -1$). A stable period-2 cycle is born. As r increases further:

- $r \approx 3$: period-2 cycle born.
- $r \approx 3.449$: period-4 cycle born.
- $r \approx 3.544$: period-8 cycle born.
- The cascade continues: period 2^n orbits for $n = 1, 2, 3, \ldots$
- The intervals between successive doublings shrink geometrically.
- At $r_\infty \approx 3.5699$: all periods 2^n coexist—the onset of chaos.

10.4.3 Period-Doubling Cascade and Feigenbaum's Universality

Let r_n be the parameter value at which the period-2^n cycle is born. Feigenbaum (1978) discovered that the ratios

$$\frac{r_n - r_{n-1}}{r_{n+1} - r_n} \to \delta \approx 4.669\,201\ldots \tag{47}$$

converge to a universal constant δ (**Feigenbaum's constant**), independent of the specific map f, as long as f has a single quadratic maximum.

10.4.4 Periodic Windows

For $r > r_\infty$, chaos is interspersed with **periodic windows**— intervals of r where a stable periodic orbit exists. The most prominent is the period-3 window near $r \approx 3.83$.

Theorem 10.1 (Li–Yorke Theorem (1975))**.** *If a continuous map on an interval has a period-3 orbit, then it has periodic orbits of* every *period.*

10.5 Lyapunov Exponent

Definition 10.3 (Lyapunov Exponent for Maps)**.** For a map $x_{n+1} = f(x_n)$ with initial condition x_0, the **Lyapunov exponent** is

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |f'(x_k)| = \lim_{n\to\infty} \frac{1}{n} \ln \left| \frac{df^n}{dx} \bigg|_{x_0} \right|. \tag{48}$$

Interpretation:

- $\lambda > 0$: nearby orbits diverge exponentially—the system is **chaotic**.
- $\lambda < 0$: nearby orbits converge—a stable periodic orbit attracts.
- $\lambda = 0$: marginal case (e.g., at a bifurcation point or neutral stability).

For two nearby initial conditions x_0 and $\tilde{x}_0 = x_0 + \delta_0$:

$$|\delta_n| \approx |\delta_0|\, e^{\lambda n} \quad \Longrightarrow \quad \lambda \approx \frac{1}{n} \ln \frac{|\delta_n|}{|\delta_0|}.$$

10.6 Renormalization

The universality of the period-doubling cascade can be understood through **renormalization group** ideas.

The key insight: near the period-doubling accumulation point r_∞, the map f^{2^n} (restricted to a small interval near the critical point) looks approximately like a rescaled copy of f itself. This self-similarity leads to a functional equation (the **Feigenbaum equation**):

$$g(x) = -\alpha\, g\big(g(-x/\alpha)\big), \tag{49}$$

where g is the universal limiting function and $\alpha \approx 2.5029$ is the rescaling factor. The constant δ (Feigenbaum's constant) arises as the relevant eigenvalue of the renormalization operator linearized about g.

11 Fractals and Strange Attractors

11.1 Fractal Dimension

Chaotic attractors typically have a **fractal** (non-integer) dimension. This reflects their infinitely detailed, self-similar structure.

Definition 11.1 (Box-Counting Dimension)**.** Cover the set S with $N(\epsilon)$ boxes of side length ϵ. The **box-counting dimension** (or Minkowski dimension) is

$$d = \lim_{\epsilon \to 0} \frac{\ln N(\epsilon)}{\ln(1/\epsilon)}. \tag{50}$$

Example 11.1 (Self-Similar Fractals)**.**

- **Cantor set:** Formed by repeatedly removing the middle third of intervals. Dimension $d = \ln 2/\ln 3 \approx 0.631$.
- **Sierpinski triangle:** Dimension $d = \ln 3/\ln 2 \approx 1.585$.
- **Koch snowflake:** Dimension $d = \ln 4/\ln 3 \approx 1.262$.

For a self-similar fractal made of N copies of itself scaled by factor r, the dimension is $d = \ln N/\ln(1/r)$.

11.2 Dimension of Self-Similar Fractals

Definition 11.2 (Self-Similarity Dimension)**.** If a set S is the union of N non-overlapping copies of itself, each scaled by a factor r (with $0 < r < 1$), then the **self-similarity dimension** is

$$d = \frac{\ln N}{\ln(1/r)}. \tag{51}$$

This coincides with the box-counting dimension for exact self-similar sets. For strange attractors (which are typically self-similar only in a statistical sense), one uses the box-counting or correlation dimension.

11.3 Strange Attractors

Definition 11.3 (Strange Attractor)**.** An attractor is **strange** if it has a fractal structure (non-integer dimension). A **chaotic attractor** is a strange attractor on which trajectories exhibit sensitive dependence on initial conditions.

Key examples:

- The **Lorenz attractor** has dimension ≈ 2.06—slightly more than a surface, reflecting its layered structure.
- The **Hénon map** $x_{n+1} = 1 - ax_n^2 + y_n$, $y_{n+1} = bx_n$ (with $a = 1.4$, $b = 0.3$) has a strange attractor with dimension ≈ 1.26.
- The **Rössler system** $\dot{x} = -y - z$, $\dot{y} = x + ay$, $\dot{z} = b + z(x - c)$ produces a simpler "band" attractor that is easier to visualize than Lorenz.

Stretching and Folding

Strange attractors arise from the interplay of two mechanisms:

1. **Stretching:** Nearby trajectories diverge (positive Lyapunov exponent), creating sensitive dependence.
2. **Folding:** Since the attractor is bounded, the stretched-out trajectories must fold back on themselves, creating the layered fractal structure.

This stretch-and-fold mechanism is analogous to kneading dough—each iteration creates finer and finer layers.

Part IV
Collective Behavior

12 The Kuramoto Model

12.1 Motivation: Synchronization in Nature

Synchronization phenomena are ubiquitous: fireflies flashing in unison, cardiac pacemaker cells, audience clapping in rhythm, circadian rhythms, and power grids. The challenge is to understand how a large population of coupled oscillators can spontaneously synchronize.

12.2 The Model

Yoshiki Kuramoto (1975) proposed a beautifully simple model. Consider N oscillators with phases $\theta_1, \ldots, \theta_N$. Each oscillator has its own natural frequency ω_i, drawn from a distribution $g(\omega)$ (typically symmetric and unimodal, e.g., Gaussian or Lorentzian). The coupling is all-to-all and sinusoidal:

$$\boxed{\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \qquad i = 1, \ldots, N.} \tag{52}$$

Here $K \geq 0$ is the **coupling strength**.

12.3 The Order Parameter

To measure the degree of synchronization, Kuramoto introduced the **complex order parameter**:

$$re^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}, \tag{53}$$

where $r \in [0, 1]$ is the **coherence** (amplitude of the mean field) and ψ is the **average phase**.

- $r = 1$: all oscillators are perfectly synchronized ($\theta_j = \psi$ for all j).
- $r = 0$: the phases are uniformly distributed around the circle (complete incoherence).

- $0 < r < 1$: partial synchronization.

Mean-field form. Multiplying both sides of (53) by $e^{-i\theta_i}$ and taking the imaginary part:

$$r \sin(\psi - \theta_i) = \frac{1}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i).$$

Thus the Kuramoto model becomes:

$$\dot{\theta}_i = \omega_i + Kr \sin(\psi - \theta_i). \tag{54}$$

Each oscillator interacts only with the "mean field" (r, ψ), not with every other oscillator individually.

12.4 Self-Consistency Analysis (Infinite N)

In the $N \to \infty$ limit, we look for a steady-state solution in a rotating frame where ψ is constant (choose a frame where $\psi = 0$). Oscillators split into two groups:

1. **Locked oscillators** ($|\omega_i| \leq Kr$): these are entrained to the mean field and satisfy $\dot{\theta}_i = 0$, i.e., $\omega_i = -Kr \sin \theta_i$, giving $\theta_i = \arcsin(\omega_i / Kr)$.
2. **Drifting oscillators** ($|\omega_i| > Kr$): these are not captured by the mean field and continue to rotate, but nonuniformly. Their steady-state density is inversely proportional to their angular velocity: $\rho(\omega, \theta) \propto 1/|\dot{\theta}|$.

The self-consistency condition requires the locked and drifting oscillators to produce exactly the assumed order parameter r. This yields:

$$r = \int_{-\pi/2}^{\pi/2} Kr \cos^2 \theta \; g(Kr \sin \theta) \, d\theta, \tag{55}$$

or equivalently, after canceling r (for $r \neq 0$):

$$1 = K \int_{-\pi/2}^{\pi/2} \cos^2 \theta \; g(Kr \sin \theta) \, d\theta. \tag{56}$$

12.5 Critical Coupling and the Phase Transition

Theorem 12.1 (Onset of Synchronization)**.** *There exists a critical coupling strength*

$$K_c = \frac{2}{\pi \, g(0)}, \tag{57}$$

such that:

- *For $K < K_c$: the only stable solution is $r = 0$ (complete incoherence).*
- *For $K > K_c$: a partially synchronized state with $r > 0$ emerges.*

Near the transition, $r \sim \sqrt{K - K_c}$, analogous to a ***supercritical pitchfork bifurcation*** *(or a second-order phase transition in statistical mechanics).*

Derivation of K_c. From (56), in the limit $r \to 0^+$:

$$1 = K \int_{-\pi/2}^{\pi/2} \cos^2\theta \; g(0)\, d\theta = K g(0) \cdot \frac{\pi}{2},$$

since $g(Kr\sin\theta) \to g(0)$ uniformly and $\int_{-\pi/2}^{\pi/2} \cos^2\theta \, d\theta = \pi/2$. Solving: $K_c = 2/(\pi g(0))$. □

Analogy with Phase Transitions

The Kuramoto model undergoes a transition remarkably similar to a **ferromagnetic phase transition**: below K_c ("high temperature"), there is no order; above K_c ("low temperature"), spontaneous order emerges. The order parameter r plays the role of the magnetization, and K plays the role of inverse temperature. The square-root scaling $r \sim \sqrt{K - K_c}$ gives a critical exponent of $\beta = 1/2$, the same as mean-field theory in statistical mechanics.

A Summary of Key Theorems

1. **Existence and Uniqueness** (1.1): Guarantees local existence and uniqueness for C^1 vector fields.
2. **Hartman–Grobman** (5.3): Hyperbolic fixed points of nonlinear systems are topologically equivalent to the corresponding linear system.
3. **Dulac's Criterion** (6.2): Rules out closed orbits via divergence conditions.
4. **Poincaré–Bendixson** (6.3): A confined trajectory in 2D must approach a closed orbit.
5. **Li–Yorke**: Period 3 implies chaos (existence of all periods).

B Useful Mathematical Facts

B.1 Taylor Expansion for Stability Analysis

For a function $f \in C^n$ near a point x^*:

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2 + \cdots + \frac{f^{(n)}(x^*)}{n!}(x - x^*)^n + O\big((x - x^*)^{n+1}\big).$$

B.2 The Jacobian Matrix

For a vector field $\dot{\vec{x}} = \vec{F}(\vec{x})$ in $\mathbb{R}^n$:

$$J_{ij} = \left.\frac{\partial F_i}{\partial x_j}\right|_{\vec{x}^*}.$$

The eigenvalues of J determine the local stability of the fixed point $\vec{x}^*$.

B.3 Green's Theorem (for Dulac's Criterion)

For a continuously differentiable vector field $\vec{A}(x, y) = (P(x, y), Q(x, y))$ and a simple closed curve C bounding a region D:

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

In divergence form (used for Dulac):

$$\iint_D \nabla \cdot \vec{A}\,dA = \oint_C \vec{A} \cdot \hat{n}\,dl.$$

B.4 Eigenvalues of a 2×2 Matrix

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:

$$\lambda = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad - bc)}}{2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}.$$

B.5 Polar Coordinate Conversions

$x = r\cos\theta$, $y = r\sin\theta$; $\quad r = \sqrt{x^2 + y^2}$, $\theta = \arctan(y/x)$.

Useful derivatives:

$$r\dot{r} = x\dot{x} + y\dot{y}, \tag{58}$$

$$r^2\dot{\theta} = x\dot{y} - y\dot{x}. \tag{59}$$